



Grade 10 Geography



3. Folding and Faulting

Learning Guide



1. Link to Plate Tectonics

- **Plate movement** causes **stress in the Earth's crust**.
- This stress can cause rocks to either:
 - **Bend** → called **folding**
 - **Break** → called **faulting**



Key idea:

Plate movement creates pressure in the crust, and this pressure changes the shape of rock layers.



2. What is Folding?

- **Folding** is the **bending of rock layers under pressure**.
- It usually happens in **softer rocks**, especially **sedimentary rocks**.
- Instead of breaking, the rock layers bend because they can handle pressure more easily.

Folding Creates:

- **Mountains**
- **Valleys**



Simple idea:

Folding = rocks bend because of pressure.



3. Factors Affecting Folding

Folding does not happen the same way everywhere. It depends on:

1. Rock Elasticity

- This means the ability of rock to **bend without breaking**.
- More elastic rocks are more likely to fold.

2. Type of Rock

- **Soft rocks** → fold easily.
- **Hard rocks** → may break instead of folding.

3. Amount of Pressure Applied

- Gentle pressure may form simple folds.
- Strong pressure may create more complex folds.



Exam tip:

Soft rock bends; hard rock breaks.



4. Basic Types of Folds

Anticline

- An **anticline** is a fold that **bends upwards**.
- It forms a raised structure.
- It is commonly linked to the formation of **mountains**.

Syncline

- A **syncline** is a fold that **bends downwards**.
- It forms a lower structure.
- It is commonly linked to the formation of **valleys**.



Memory tip:

Anticline = up
Syncline = down

5. Symmetrical and Asymmetrical Folds

Symmetrical Fold

- Both sides of the fold are **equal**.
- The fold looks balanced.
- This usually happens when pressure is applied more evenly.

Asymmetrical Fold

- One side is **steeper** than the other.
- The fold is uneven.
- It is caused by **unequal pressure**.

✚ Key idea:

Equal pressure forms more balanced folds, while unequal pressure forms uneven folds.

6. Classification of Folds

Folds can be grouped into:

- **Symmetrical folds**
- **Asymmetrical folds**

Examples of Folds

- **Chevron folds**
- **Isoclinal folds**

These are more complex examples of folded rock layers.

7. Advanced Types of Folding

Open Fold

- Has a **wide angle between limbs**.
- Forms under **low to moderate compressional forces**.
- The rocks are **gently folded**, not tightly squeezed.
- Common in areas with less intense tectonic activity.

✚ Simple idea:

Open folds show weaker pressure.

Overfold

- One limb is **pushed over the other**.
- Caused by **strong compressional forces**.
- Both limbs dip in the **same direction**.

✚ Simple idea:

Overfolds show stronger pressure than open folds.

Overthrust Fold

- The fold becomes **fractured or faulted**.
- It is then **pushed forward**.
- Movement happens along a **thrust fault**.
- Older rocks may be pushed over younger rocks.

✚ Important:

Overthrust folding combines both folding and faulting.

Chevron Fold

- A chevron fold has **sharp, angular folds**.
- It has **straight limbs**.
- It forms under **strong pressure** in layered rocks.

✚ **Simple idea:**

Chevron folds look sharp and zigzag-like.

⚠ **8. What is Faulting?**

- **Faulting** is the **breaking of rock layers under pressure**.
- It usually occurs in **hard, brittle rocks**.
- These rocks cannot bend easily, so they break.

Faulting Forms:

- **Fault lines**
- **Fault planes**

✚ **Key idea:**

Faulting = rocks break because of pressure.

✚ **9. Key Faulting Terms**

Fault Line

- A **line of weakness** in the Earth's crust.
- Movement may occur along this line.

Fault Plane

- The **surface where movement occurs**.
- It is the actual break or fracture surface.

Hanging Wall

- The rock block **above the fault plane**.

Footwall

- The rock block **below the fault plane**.

✚ **Memory tip:**

Hanging wall is above; footwall is below.

🔄 **10. Types of Faults**

1. Normal Fault

- Caused by **tension**, meaning rocks are being **pulled apart**.
- The **footwall is raised** in relation to the hanging wall.
- It is a **vertical fault**.

✚ **Key idea:**

Normal fault = pulling apart.

2. Reverse Fault

- Caused by **compression**, meaning rocks are being **pushed together**.
- The **hanging wall is raised** in relation to the footwall.
- It is also a **vertical fault**.

✚ **Key idea:**

Reverse fault = pushing together.

3. Strike-Slip / Tear Fault

- Caused by **sideways movement**.
- The blocks move horizontally past each other.
- It is a **horizontal fault**.

✚ **Key idea:**

Strike-slip fault = sideways movement.

🔄 **11. Movement Along Faults**

Faults can move in different ways:

- **Vertically** → up and down movement.
- **Horizontally** → sideways movement.

- **Combination movement** → also called **compound faults**.

📌 **Simple idea:**

Fault movement can be vertical, horizontal, or both.

12. Landforms from Faulting

1. Block Mountain / Horst

- A **horst** is a **raised block between faults**.
- It forms when part of the Earth's crust is pushed above its surroundings.
- It is bordered by faults on both sides.
- Example: **Ruwenzori Mountains, East Africa**.

📌 **Memory tip:**

Horst = raised block.

2. Rift Valley / Graben

- A **graben** is a **sunken block between faults**.
- It forms when a crustal block sinks below its surroundings.
- It is bordered by faults on both sides.
- Example: **Great Rift Valley, Africa**.

📌 **Memory tip:**

Graben = sunken block.

13. Final Exam Summary

- Plate movement causes **stress in the Earth's crust**.
 - **Folding** happens when rocks **bend**.
 - Folding usually occurs in **soft rocks**.
 - Folding forms **mountains and valleys**.
 - **Faulting** happens when rocks **break**.
 - Faulting usually occurs in **hard, brittle rocks**.
 - **Anticline** bends upward → mountains.
 - **Syncline** bends downward → valleys.
 - **Normal fault** is caused by **tension**.
 - **Reverse fault** is caused by **compression**.
 - **Strike-slip fault** is caused by **sideways movement**.
 - Faulting forms:
 - **Horsts / block mountains**
 - **Grabens / rift valleys**
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